THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Why can we not see our image in a sheet of paper? 1

(a) paper is transparent **(b)** diffused reflection
(c) paper absorbs all **(d)** regular reflection

2. If two mirrors are parallel, the number of images formed is (in theory): 1

(a) 1 **(b)** very large **(c)** 4 **(d)** 2

3. A pinhole camera of length 20 cm views a 2 m tall object at 10 m. The image height is: 1

(a) 40 cm **(b)** 2 cm **(c)** 20 cm **(d)** 4 cm

4. A calm pond shows an inverted image of a tree. This is due to: 1

(a) absorption **(b)** diffused reflection
(c) refraction **(d)** regular reflection

5. A bright object is 2 m from a pinhole and the screen is 10 cm behind. The object is 20 cm tall. The image height is: 1

(a) 2 cm **(b)** 0.1 cm **(c)** 1 cm **(d)** 10 cm

6. If the object distance is halved, the image height in the above becomes: 1

(a) same **(b)** half **(c)** quarter **(d)** double

7. Which of these is a correct statement about shadows? 1

(a) a shadow is formed by a transparent object
(b) a shadow is formed without light
(c) a shadow is a coloured object
(d) a shadow is darker where no light reaches

8. A man looks at his image in a mirror and sees AMBULANCE reading correctly on the car in front through the mirror. The car's lettering is: 1

(a) upside down **(b)** normal
(c) faint **(d)** laterally inverted

9. **Assertion (A):** A pinhole camera forms an inverted image of an object. 1
Reason (R): Light travels in straight lines and the rays cross at the pinhole.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** In reflection from a plane mirror, the angle of incidence equals the angle of reflection. 1

Reason (R): The incident ray, the reflected ray and the normal lie in the same plane.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why the writing on a vehicle's front is reversed. 2

12. Why is the shadow of a ball sharp under a small lamp but blurred under a tube-light? 2

13. Why is it important to keep roads clear of glare reflecting surfaces? 2

14. How could a teacher use two mirrors to show the number of images varies with the angle? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A small lamp casts a shadow of a book 12 cm wide into a screen. The book is 6 cm wide and 20 cm from the lamp. Find the distance of the screen from the lamp. 3

16. Explain the working of a simple periscope made from a cardboard tube and two small mirrors. 3

17. Describe a simple activity to find the position of image using two pins and a mirror. 3

OR

Explain why the sun's image formed by a pinhole appears circular even if the hole is square. 3

18. What happens to the angle of reflection if the incident ray is rotated by 20° about the point of incidence? 3

OR

Describe the effect of moving a screen away from the pinhole on the brightness and size of the image. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sameer makes a pinhole camera using a cardboard tube, with a tiny hole at one end and a tracing paper screen at the other end. He points the hole at a burning candle and sees an image on the screen. 5

(i) Is the image on the screen erect or inverted?

(ii) Is the image real or virtual?

(iii) What will happen if the hole is made larger?

(iv) Why is the image inverted?

20. Explain why the length of a person's shadow is greatest in the early morning and late evening. 5

OR

Write a short project report on 'Light and its reflections in my home' with observations and examples. 5

END OF QUESTION PAPER — Total: 40 Marks